

SPOKANE INTL, WA, USA

WMO: 727850

Lat: 47.622N

Lon: 117.528W

Elev: 717

StdP: 93.00

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24157

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-14.4	-11.2	-19.4	0.7	-12.7	-16.0	1.0	-9.2	13.6	4.5	11.9	3.9	2.4	30	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	34.0	17.0	32.3	16.4	30.3	15.8	18.3	30.5	17.5	29.5	16.6	28.1	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	11.1	20.2	13.1	10.2	19.9	12.0	9.5	19.5	54.2	30.5	51.3	29.4	48.8	28.1	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.7	8.5	-18.9	37.0	4.4	1.6	-22.1	38.2	-24.7	39.1	-27.2	40.0	-30.4	41.2		
(5)			-19.3	19.8	4.3	1.5	-22.4	20.8	-25.0	21.7	-27.4	22.5	-30.5	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-1.4	0.2	4.3	8.1	13.1	16.6	21.8	21.2	16.0	8.6	2.4	-1.8		
	DBStd	9.16	4.66	4.40	3.63	3.62	4.21	3.96	3.66	3.57	4.11	3.89	4.32	4.43		
	HDD10.0	1576	353	276	178	80	17	2	0	0	4	71	229	367		
	HDD18.3	3633	612	508	435	308	170	78	12	14	92	301	478	625		
	CDD10.0	1264	0	0	2	23	113	201	366	346	184	29	1	0		
	CDD18.3	281	0	0	0	0	8	27	120	102	23	1	0	0		
	CDH23.3	3626	0	0	0	7	119	373	1545	1266	310	6	0	0		
	CDH26.7	1511	0	0	0	1	26	124	710	551	99	1	0	0		
(14) Wind	WSAvg	3.8	3.5	3.7	4.3	4.3	4.0	4.0	3.7	3.5	3.4	3.6	3.7	3.5		
(15) Precipitation	PrecAvg	397	47	35	37	30	38	29	15	16	17	26	53	56		
	PrecMax	641	114	83	97	65	93	72	59	48	48	83	122	125		
	PrecMin	267	10	1	6	2	5	3	0	0	0	1	5	14		
	PrecStd	74	23	19	18	17	21	16	16	14	13	20	29	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	11.5	17.3	24.0	29.6	33.5	36.7	35.9	32.5	23.8	14.6	10.3		
		MCWB	6.4	7.5	9.2	12.7	14.9	17.1	18.2	17.4	16.0	13.3	11.2	7.9		
	2%	DB	6.8	9.2	14.4	20.6	26.7	30.0	34.4	33.8	29.6	20.0	11.8	7.2		
		MCWB	4.9	6.1	8.1	11.4	14.3	15.9	17.2	16.9	15.0	11.6	8.9	5.4		
	5%	DB	5.2	7.3	12.3	17.6	24.0	27.4	32.7	32.0	27.0	17.3	9.9	5.1		
		MCWB	3.9	4.8	7.2	9.7	13.0	15.1	16.6	16.2	14.3	10.6	7.3	3.8		
	10%	DB	3.9	5.7	10.3	15.1	21.4	25.1	30.9	29.9	24.4	14.9	8.3	3.6		
		MCWB	2.8	3.5	6.0	8.3	12.0	14.1	16.1	15.7	13.5	9.2	6.2	2.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.4	10.7	14.1	16.4	18.3	20.0	19.1	17.4	14.5	11.7	8.0		
		MCDB	8.3	10.4	14.9	21.9	25.8	30.6	31.8	32.0	27.9	21.0	13.4	9.8		
	2%	WB	5.2	6.6	8.9	11.8	15.0	16.9	18.7	18.0	16.1	12.6	9.3	5.6		
		MCDB	6.4	8.6	13.1	18.6	24.0	27.5	30.6	30.6	27.0	18.2	11.4	7.0		
	5%	WB	4.1	5.2	7.7	10.2	14.0	15.8	17.8	17.1	15.1	11.3	7.8	4.1		
		MCDB	5.0	7.0	11.5	16.7	22.2	25.5	30.0	29.4	25.2	16.2	9.4	4.9		
	10%	WB	2.9	3.8	6.6	9.0	12.8	14.9	16.8	16.3	14.0	10.0	6.6	2.6		
		MCDB	3.7	5.4	10.1	14.2	19.9	23.8	28.6	28.2	23.1	14.0	8.2	3.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	7.1	9.1	10.6	11.8	12.2	14.5	14.4	13.2	10.3	6.7	5.1		
		MCDBR	6.0	8.6	12.1	14.2	15.3	15.5	16.7	16.9	16.8	13.5	8.1	5.9		
	5% WB	MCWBR	4.6	5.8	7.2	7.1	6.2	5.6	5.7	6.2	6.7	7.0	5.6	4.9		
		MCWBR	5.7	7.6	10.6	13.0	13.5	14.2	15.4	15.4	15.1	11.9	7.1	5.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.272	0.280	0.294	0.312	0.325	0.327	0.327	0.337	0.314	0.300	0.280	0.268		
		taud	2.499	2.539	2.555	2.490	2.472	2.496	2.496	2.452	2.507	2.544	2.556	2.501		
	Ebn at Noon	Ebn at Noon	832	903	938	945	938	934	930	906	901	863	816	790		
		Edh at Noon	62	73	85	99	105	104	102	102	88	72	58	55		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.04	1.97	3.03	4.53	5.83	6.27	7.38	6.12	4.36	2.59	1.23	0.82		
	RadStd	RadStd	0.13	0.27	0.36	0.43	0.47	0.55	0.40	0.40	0.41	0.29	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+51	
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	-0.40	N/A	N/A	+95	+44	

Nomenclature: See separate page